Fourier Phase Preservation and Spectral Random Augmentation for Cross-Scene Hyperspectral Image Classification With Ensemble Networks

Kai Yang, Boshan Shi, Guo Cao, *Member, IEEE*, and Youqiang Zhang, *Member, IEEE*

***Abstract*—Cross-scene hyperspectral image (HSI) classification faces significant challenges due to spectral–spatial domain shifts. Existing domain generalization (DG) methods typically generate extended domains (EDs) to enhance the diversity of source domains (SDs) and extract domain-invariant features for target domain (TD) classification. However, these methods often struggle to balance the invariance and diversity of ED features. For instance, placing excessive emphasis on diversity can compromise cross-domain invariance. Additionally, they tend to overlook domain-specific features that are crucial for classification, which limits generalization performance. To address these challenges, this article proposes an ensemble network based on Fourier phase preservation and spectral random augmentation (FSENet), which generates high-quality EDs and effectively integrates domain-invariant and domain-specific features to improve classification accuracy. FSENet consists of three key components: 1) the Fourier Spatial Augmentation Module (FSAM), which preserves phase while perturbing amplitude to enhance both the invariance and diversity of spatial features; 2) the Random-augmented Spectral Channel Attention Module (RSCAM), which employs channel attention and random perturbations to boost the discriminative ability of spectral features; and 3) the Domain Label-guided Ensemble Framework (DLEF), which utilizes a multibranch architecture to combine domain-invariant and domain-specific features from multiple SDs, overcoming the limitations of single-feature representations. Experiments on four real-world hyperspectral datasets, including Houston, Pavia, HyRANK, and WHU, demonstrate that FSENet achieves a cross-scene classification accuracy improvement of 2.27%–4.74% compared to state-of-the-art domain adaptation (DA) and DG methods. Ablation studies confirm the effectiveness of each proposed module. The code of this work will be available at https://github.com/goesfor/FSENet**

***Index Terms*—Channel attention mechanisms, ensemble learning, Fourier-based augmentation, hyperspectral image (HSI) classification, single-source domain generalization (DG).**

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I. Introduction

In RECENT years, hyperspectral image (HSI) classification has been widely applied across various fields, including crop monitoring [1], [2], urban development [3], [4], mineral exploration [5], [6], and environmental resource management [7], [8], [9]. The objective of HSI classification is to categorize pixels into distinct classes based on their spectral and spatial features. Although mainstream deep learning techniques demonstrate exceptional performance [10], [11], [12], their training typically requires a large volume of accurately labeled samples. Acquiring sufficient labeled data for new scenarios can be time-consuming and resource-intensive. Furthermore, models trained using these methods often struggle to generalize to previously unseen situations. Consequently, enhancing the generalization ability of models in unknown environments has become a key area of research.

Spectral drift [13], [14] and spatial texture variation, caused by differences in sensor types, diurnal and seasonal fluctuations, and weather conditions, lead to inconsistencies in data distributions between the source domain (SD) and target domain (TD). When models trained on SD are applied to TD, this inconsistency significantly degrades performance. To tackle this issue, domain adaptation (DA) methods [15] have been widely used for cross-scenario HSI classification. Although DA methods produce promising classification results, they require TD data for training, which limits the applicability of the resulting models to specific target scenarios. To address these limitations, domain generalization (DG) methods have gained increasing attention, as they require only SD data for training and can be directly applied to unseen TD.

DG has been effectively utilized in various fields, including medical image analysis [16], fault detection [17], and natural language processing [18]. Recently, DG in HSI classification has garnered significant research attention. DG can be categorized into multisource and single-source types, depending on the number of SDs involved. Qi et al. [19] proposed a Multisource DG Two-Branch Network (MDGTnet), which aligns the feature distributions of two SDs to extract domain-invariant features for TD classification. However, obtaining labeled HSI data for multiple scenes within the same category is challenging, making single-source DG more practical. Existing single-source DG methods expand the SD by generating extended domains (EDs), but

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they struggle to balance feature invariance with diversity. For example, SDEnet [20] generates EDs by applying spatial and spectral randomization in the semantic encoder and employs morphological operations in the morphological encoder to extract domain-invariant templates. However, simple randomization and morphological operations may distort semantics. Locally Linear Unbiased Randomization Network (LLURnet) [21] utilizes a symmetric encoder-random decoder architecture and 3-D convolutions to extract joint spectral–spatial features for generating EDs. This approach lacks a mechanism to preserve domain-invariant features and overlooks the significance of spectral bands, potentially introducing redundant noise. S2AMSnet [22] generates EDs through spatial and spectral variations while enforcing semantic consistency using a multiscale mutual information regularization network (MMIRN) [22]. However, this implicit constraint does not guarantee accurate preservation of domain-invariant features. S2Ecnet [23] generates spectral and spatial features using Adaptive Instance Normalization (AdaIN) and STN but fails to effectively mitigate spectral redundancy interference. Additionally, spatial transformations via STN may lead to the loss of spatial details. FDGNet [24] separates high- and low-frequency components using a frequency-decoupled spectral–spatial encoder to maintain semantic consistency; however, fixed frequency decomposition may discard critical spatial details, resulting in excessive sample bias. Furthermore, existing methods primarily focus on extracting domain-invariant features from multiple SDs and applying them directly to the TD, neglecting the crucial role of domain-specific features in classification.

To address these limitations, we propose the ensemble network based on Fourier phase preservation and spectral random augmentation (FSENet) for cross-scenario HSI classification. FSENet consists of a generator and a discriminator. The generator includes three modules: Spatial Feature Enhancement module. Inspired by Xu et al. [25], who demonstrated that amplitude components in the frequency domain capture low-level statistics, while phase components preserve cross-domain semantic invariance, we designed a Fourier spatial augmentation module (FSAM). FSAM transforms spatial features into the frequency domain, retains phase components to explicitly preserve semantic information, and perturbs amplitude components to enhance diversity, generating spatially invariant yet diverse features. Spectral Feature Enhancement module: We developed a Random-augmented Spectral Channel Attention Module (RSCAM), which first introduces a channel attention module (CAM) to adaptively assign weights to spectral bands based on the relationship between the central pixel and its neighborhood. Subsequently, randomization enhancement is applied to important bands via AdaIN, reducing the impact of redundant bands and ensuring cross-domain invariance of spectral features. Feature fusion module: A learnable weighting mechanism integrates enhanced spectral features, spatial features, and SD features to generate EDs that balance feature invariance and diversity. The discriminator primarily comprises a Domain Label-guided Ensemble Framework (DLEF). By jointly optimizing the domain classification loss and contrastive learning loss, DLEF can simultaneously extract both domain-invariant and domain-specific features. By assigning a base learner to each domain and leveraging domain similarity probabilities—derived from the domain classifier's output—as weighting factors for feature extraction, DLEF adaptively fuses features from all SDs. This enables more accurate classification in the TD.

TABLE I
CLASSES AND SAMPLE NUMBERS IN THE HOUSTON 13 AND HOUSTON 18 DATASETS

No.	Class Name	Houston13(SD)	Houston18(TD)
1	Grass healthy	345	1353
2	grass stressed	365	4888
3	Trees	365	2766
4	Water	285	22
5	Residential buildings	319	5347
6	Non-residcntial buildings	408	32459
7	Road	443	6365
	Total	2530	53200

The main contributions of this article are as follows.

1) To tackle the challenge of spatial feature generation in DG for HSI classification, we present the first Fourier-enhanced approach. To reduce the loss of domain-invariant features during ED generation, we developed the FSAM. This module separates spatial features into phase components, which preserve cross-domain semantic invariance, and amplitude components, which enhance diversity through random perturbations. This effectively minimizes feature distortion caused by conventional spatial style transfer.

2) To address the degradation of domain-invariant features caused by spectral redundancy, we proposed a novel spectral enhancement strategy: the RSCAM. This module combines channel weighting with random perturbations, dynamically adjusting the importance of different bands and applying random enhancements to balance feature invariance and diversity in spectral feature learning.

3) To overcome the limitations of relying solely on domain-invariant features, we proposed the DLEF, the first ensemble framework designed for DG in HSI classification. This framework significantly improves classification accuracy by jointly optimizing domain classification loss and contrastive learning loss, facilitating the collaborative fusion of domain-invariant and domain-specific features.

4) FSENet achieves significantly higher overall accuracy (OA) than state-of-the-art methods (e.g., +2.41% on Houston versus S2AMSnet, +2.27% on WHU versus S2AMSnet) across four real-world datasets, with detailed metrics in Tables V–VIII. This demonstrates its superior cross-scenario generalization in complex environments.

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TABLE II
CLASSES AND SAMPLE NUMBERS IN THE PAVIA UNIVERSITY AND PAVIA CENTER DATASETS

No.	Class Name	Pavia University (SD)	Pavia Center (TD)
1	Tree	3064	7598
2	Asphalt	6631	9248
3	Brick	3682	2685
4	Bitumen	1330	7287
5	Shadow	947	2863
6	Meadows	18649	3090
7	Bare soil	5029	6584
	Total	39332	39355

TABLE III
CLASSES AND SAMPLE NUMBERS IN THE DIONI AND LOUKIA DATASETS

No.	Class Name	Dioni(SD)	Loukia(TD)
1	Dense Uraban Fabric	1262	206
2	Mineral Extraction Sites	204	54
3	Non Irrigated Arable Land	614	426
4	Fruit Trees	150	79
5	Olive Groves	1768	1107
6	Coniferous Forest	361	422
7	Dense Sclerophyllous Vegetation	5035	2996
8	Sparce Sclerophyllous Vegetation	6374	2361
9	Sparsely Vegetated Areas	1754	399
10	Rocks and Sand	492	453
11	Water	1612	1393
12	Coastal Water	398	421
	Total	20024	10317

TABLE IV
CLASSES AND SAMPLE NUMBERS IN THE HONGHU AND HANCHUAN DATASETS

No.	Class Name	HongHu(SD)	HanChuan(TD)
1	Tree	4040	15558
2	Bare soil	21821	9116
3	Red roof	14041	10516
	Total	39902	35190

The remainder of this article is organized as follows. Section II reviews related work, focusing on frequency-domain theory and attention mechanisms in cross-domain classification. Section III presents the architecture of FSENet, detailing the phase–amplitude decoupling principle of FSAM, the spectral band weighting strategy of RSCAM, and the multibranch feature fusion mechanism of the DLEF. Section IV evaluates the proposed method on four cross-scenario HSI datasets, providing quantitative comparisons with eight state-of-the-art methods [e.g., OA, Kappa coefficient (KC)], a visual quality analysis of generated samples, sensitivity studies on parameters (e.g., training spectral block size, loss function weight coefficients), ablation experiments for each module (FSAM, RSCAM, DLEF), and an analysis of time complexity. Finally, Section V summarizes the three key contributions of this article and discusses the framework's limitations and future research directions.

II. RELATED WORK

A. Fourier-Based Enhancement

By disturbing the amplitude, the model can focus more on phase information, thus achieving Fourier-based data enhancement. Xu et al. [25] presented a theoretical perspective that explains why amplitude disturbance allows the model to prioritize phase information. For simplicity, we explore the scenario in which a linear classifier is paired with a feature extractor $\mathbf{h}$. Under the assumption that the Fourier-based augmented distribution is $g \sim G$, we define the training risk on the enhanced data as follows:

$$\hat{R}_{fa} = \frac{1}{N}\sum_{n=1}^{N} \mathbb{E}_{g\sim G}\left[\ell\left(\mathbf{W}^{\mathrm{T}}\mathbf{h}\left(g\left(x\right)\right), y\right)\right]. \tag{1}$$

Following [26] and [27], we use Taylor expansion to reframe (1):

$$\begin{aligned}&\mathbb{E}_{g\sim G}\left[\ell\left(\mathbf{W}^{\mathrm{T}}\mathbf{h}\left(g\left(x\right)\right), y\right)\right]\\ &\quad= \ell\left(\mathbf{W}^{\mathrm{T}}\bar{\mathbf{h}}, y\right) + \frac{1}{2}\mathbb{E}_{g\sim G}\left[\Delta^{\mathrm{T}}\mathbf{H}\left(\tau, y\right)\Delta\right]\end{aligned} \tag{2}$$

where $\bar{\mathbf{h}} = \mathbb{E}_{g\sim G}[\mathbf{h}(g(x))], \Delta = \mathbf{W}^{\mathrm{T}}(\bar{\mathbf{h}} - \mathbf{h}(g(x)))$, and $\mathbf{H}$ represents the Hessian matrix. In the case of a cross-entropy loss function utilizing a SoftMax activation function, $\mathbf{H}$ is positive semidefinite and unrelated to y. Accordingly, in order to minimize the quadratic term in (2), the weight $w_{i,d}$ assigned to features h_d exhibiting high variance $h_d(g(x))$ would be close to 0. It is assumed that h_a and h_p represent the features derived from amplitude and phase information, respectively. Because only the amplitude information changes and the phase information remains unaltered, the following equation can be derived:

$$\begin{cases}\left|h_p\left(g\left(x\right)\right) - h_p\left(x\right)\right| < \zeta\\ \left|h_a\left(g\left(x\right)\right) - h_a\left(x\right)\right| < \varepsilon\end{cases} \tag{3}$$

where $\zeta > 0$ is a small value, and $\varepsilon \geq \zeta$. As a result of the disturbance, the variance of the amplitude-dependent feature h_a is significant. Consequently, minimizing $\hat{R}_{fa}$ involves reducing the weight $w_{i,a}$ associated with h_a to a value close to 0. Therefore, the classifier would accord greater significance to the phase-dependent features h_p with comparatively greater weight when making decisions.

B. Channel Attention Mechanism

The CAM leverages the relationships among feature channels to generate channel attention maps, treating each channel in the feature map as a feature detector [28]. This approach

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TABLE V

CLASSIFICATION RESULTS (MEAN ± STANDARD DEVIATION) OF FSENET AND COMPARISON METHODS ON THE TARGET SCENE HOUSTON 18 DATASET

Class	Classification Methods								
	DSAN[44]	MRAN[45]	TSTnet[46]	SDEnet[20]	LLURnet[21]	S2AMSnet[22]	S2ECnet[23]	FDGnet[24]	FSENet
1	**67.48**	67.41	66.89	39.42	42.68	46.48	19.14	40.16	43.96
2	69.23	48.38	77.59	75.23	75.3	78.87	**85.21**	80.37	80.85
3	74.91	76.61	**77.62**	63.92	60.34	54.06	47.43	62.73	63.49
4	95.45	100	100	100	100	93.64	100	100	**100**
5	62.37	45.2	71.69	76.12	76.56	75.58	69.35	76.22	**83.85**
6	77.53	89.91	81.41	78.89	82.57	**92.34**	87.93	87.01	91.39
7	39.5	20.24	46.54	53.56	54.85	46.94	48.75	50.52	**59.93**
OA(%)	70.31±1.16	71.76±2.36	75.35±1.86	73.47±3.48	75.81±1.40	80.83±0.84	77.27±1.35	78.50±1.64	**83.24**±0.47
Kappa	52.64±1.65	49.32±1.77	60.48±2.03	57.66±4.11	60.16±2.19	66.64±2.03	61.54±2.51	63.58±2.70	**71.43**±0.71

TABLE VI

CLASSIFICATION RESULTS (MEAN ± STANDARD DEVIATION) OF FSENET AND COMPARISON METHODS ON THE TARGET SCENE PAVIA CENTER DATASET

Class	Classification Methods								
	DSAN[44]	MRAN[45]	TSTnet[46]	SDEnet[20]	LLURnet[21]	S2AMSnet[22]	S2ECnet[23]	FDGnet[24]	FSENet
1	**92.79**	72.62	80.52	88.21	89.38	91.96	87.14	87.53	89
2	82.35	83.73	79.73	80.99	86.04	82.93	78.15	81.39	**91.79**
3	49.31	64.99	70.88	69.7	60.32	64.89	47.56	72.34	**86.48**
4	77.41	47.98	70.62	83.34	77.4	82.98	**85.1**	84.49	83.08
5	86.87	86.62	88.26	89.8	87.89	83.25	93.01	86.94	**92.89**
6	74.43	**91.04**	82.52	76.08	72.58	73.92	79.35	72.17	77.48
7	67.66	70.64	67.22	74.01	78.87	81.64	**90.95**	79.38	86.19
OA(%)	78.45±0.85	70.05±1.42	76.34±1.68	81.14±1.74	81.21±1.25	82.55±0.95	82.40±1.40	81.88±2.16	**87.29**±0.34
Kappa	73.98±1.66	62.55±1.93	71.74±2.33	77.38±2.05	77.37±1.50	79.01±1.09	78.84±1.67	78.26±2.60	**84.71**±0.39

TABLE VII

CLASSIFICATION RESULTS (MEAN ± STANDARD DEVIATION) OF FSENET AND COMPARISON METHODS ON THE TARGET SCENE LOUKIA DATASET

Class	Classification Methods								
	DSAN[44]	MRAN[45]	TSTnet[46]	SDEnet[20]	LLURnet[21]	S2AMSnet[22]	S2ECnet[23]	FDGnet[24]	FSENnet
1	26.7	16.02	26.21	28.2	30.29	10.53	27.18	**41.26**	0.05
2	27.78	37.04	1.85	29.07	2.78	0	44.44	12.96	**96.48**
3	41.55	32.39	53.05	44.79	46.74	38.54	**54.23**	35.92	0.42
4	8.86	**11.39**	0	10.38	5.06	7.6	5.06	5.06	0.25
5	0.36	**5.15**	0.18	3.34	0.37	0.18	0.36	0.09	0.02
6	41.23	**48.58**	41	39.12	36.54	26.33	44.55	43.36	32.06
7	77.2	70.26	76.17	76.29	76.37	80.19	70.96	73.13	**85.63**
8	58.66	65.44	70.31	63.49	68.42	69.82	70.82	72.55	**76.99**
9	72.68	**73.93**	30.08	57.57	30.78	24.46	57.14	28.32	69.07
10	0	3.09	18.54	14.73	17.92	0	0	37.31	**51.38**
11	100	100	100	99.98	100	100	100	100	**100**
12	100	100	100	100	99.95	99.31	100	**100**	98.79
OA(%)	60.42±3.07	60.43±2.34	62.19±1.52	61.71±1.13	61.52±1.11	60.72±0.64	61.52±0.78	62.37±0.47	**66.78**±0.50
Kappa	51.12±2.89	51.46±2.11	52.92±1.78	52.74±1.35	52.18±1.27	50.04±1.24	52.51±1.03	53.32±0.51	**58.07**±0.80

effectively identifies the most important parts of the input image. To ensure efficient computation of channel attention, it is essential for the CAM to compress the spatial dimensions of the feature maps. Average pooling is commonly used to aggregate spatial information. Zhou et al. [29] recommended this method for effectively learning the extent of target objects, while Hu et al. [30] employed it to reduce the spatial dimensions of feature maps in their attention module. Building on these studies, Woo et al. [31] argued that max pooling can capture additional unique features of objects and help infer fine-grained channel attention. Consequently, CAM combines average pooling features with max pooling features, significantly enhancing the network's representativeness. The configuration of CAM is illustrated in Fig. 1, and the process

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TABLE VIII
CLASSIFICATION RESULTS (MEAN ± STANDARD DEVIATION) OF FSENET AND COMPARISON METHODS ON THE TARGET SCENE HANCHUAN DATASET

Class	Classification Methods								
	DSAN[44]	MRAN[45]	TSTnet[46]	SDEnet[20]	LLURnet[21]	S2AMSnet[22]	S2ECnet[23]	FDGnet[24]	FSENet
1	94.92	**96.94**	96.64	92.86	92.2	95.02	94.32	94.89	95.85
2	81.77	59.65	68.2	85.23	**88.74**	80.65	88.6	83.45	80.21
3	23.56	34.22	33.82	29.96	39.67	63.86	30.44	38.52	**70.94**
OA(%)	71.78±3.39	70.36±2.28	72.18±1.98	73.42±1.22	76.67±3.67	82.82±4.85	75.07±3.12	76.36±1.29	**85.09**±1.97
Kappa	55.80±2.89	52.42±1.93	55.69±2.35	58.61±2.05	63.85±5.56	72.90±7.54	61.22±4.18	62.97±2.02	**76.41**±3.24

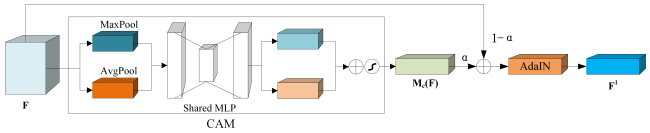

Fig. 1. Schematic of RSCAM.

for calculating channel attention is detailed below

$$M_c(\mathbf{F}) = \sigma\left(\mathrm{MLP}\left(\mathrm{AvgPool}(\mathbf{F})\right) + \mathrm{MLP}\left(\mathrm{MaxPool}(\mathbf{F})\right)\right) = \sigma\left(\mathbf{w_1}\left(\mathbf{w_0}\left(\mathbf{F}^{\mathbf{c}}_{\mathbf{avg}}\right)\right) + \mathbf{w_1}\left(\mathbf{w_0}\left(\mathbf{F}^{\mathbf{c}}_{\mathbf{max}}\right)\right)\right) \quad (4)$$

where $\boldsymbol{\sigma}$ represents the sigmoid function, $\mathbf{W_0} \in \mathbb{R}^{C/r\times C}$, and $\mathbf{W}_1 \in \mathbb{R}^{C\times C/r}$. It should be noted that both inputs share the weights $\mathbf{W_0}$ and $\mathbf{W}_1$ of the MLP, and the ReLU activation function applied subsequent to $\mathbf{W_0}$.

C. Ensemble Learning

Ensemble learning is a machine learning technique that aims to enhance generalization performance by combining multiple learners rather than relying on a single one [32]. This approach is based on the premise that the collective performance of a group of learners often exceeds that of any individual learner [33]. Common ensemble learning methods primarily fall into two categories: Boosting and Bagging. Boosting organizes individual learners sequentially [34], whereas Bagging arranges them in parallel [35]. This article hypothesizes that different domains share certain similarities, allowing the TD to be represented as a linear combination of multiple SDs. Therefore, we adopt the Bagging method, setting up an individual learner for each SD to extract features in parallel. Furthermore, by introducing domain classification, we measure the similarity between the TD and each SD, using these similarities as weights to combine the features extracted by each learner for the final classification.

III. PROPOSED METHOD

In single-source DG methods, the model is trained exclusively on SD data without access to TD samples. To ensure robust performance upon transfer to the TD, FSENet employs a generator architecture with two branches: one dedicated to spatial feature enhancement and the other to spectral feature enhancement. This dual-branch design facilitates the generation of high-quality ED data, thereby effectively improving the model's generalization capability.

Specifically, in the spatial feature enhancement branch, to mitigate the loss of domain-invariant features often caused by conventional style transfer techniques, we draw inspiration from well-established theories in natural image processing. These theories suggest that phase information more effectively preserves domain-invariant features. By applying the Fourier Transform to HSI, we enhance spatial features from a frequency domain perspective. Specifically, we decompose the images into phase and magnitude components, retaining the phase component while applying targeted transformations to the magnitude component. The modified data is then transformed back to the spatial domain, yielding enhanced spatial features. In the spectral feature enhancement branch, to mitigate noise issues caused by redundant spectral bands, we proposed the RSCAM. This module first identifies information-rich spectral bands and then applies random enhancement, resulting in improved spectral feature representation. At the final stage of the generator, a learnable weighting mechanism combines spatial features, spectral features, and the SD features to derive an ED. To ensure that the ED does not diverge significantly from the SD, a mutual information (MI) loss [22] is introduced. Additionally, to facilitate more stable learning of domain-invariant features by the discriminator, the ED and SD are blended at a learnable ratio to form a mixed domain (MD). The SD, ED, and MD are then fed into the discriminator. After passing through a feature extractor to obtain intermediate representations, supervised contrastive learning is applied to enforce domain-invariant feature extraction. Additionally, a domain classification module is incorporated, leveraging a domain classification loss. This loss serves a dual purpose: preventing the generator from producing ED samples that are overly similar to the SD and ensuring that the intermediate features retain domain-specific characteristics. Additionally, an ensemble learning approach is introduced, where a distinct base learner is assigned to each SD. Each base learner extracts both domain-invariant and domain-specific features from its respective SD. The output probabilities from the domain classification module quantify the similarity between the TD and each SD, serving as weights for feature fusion across multiple SDs. This weighted fusion improves the classification accuracy of TD. The architecture of FSENet, illustrated in Fig. 2, comprises a generator and a discriminator. The generator includes a spatial feature enhancement module, a spectral feature enhancement module, and a feature fusion module. The discriminator consists of a

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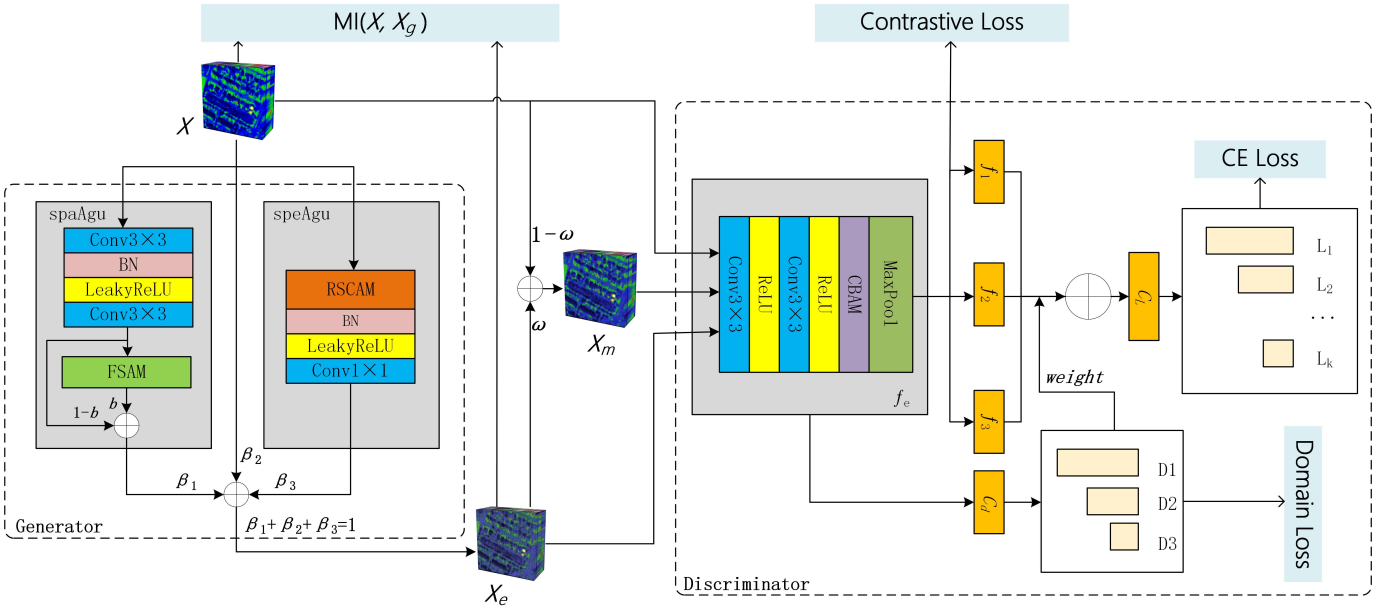

Fig. 2. Schematic of FSENet for cross-scene HSI classification. The generator enhances spatial and spectral features through a dual-branch architecture to generate EDs. The discriminator first extracts shallow features from multiple SDs, then performs multibranch feature fusion guided by domain labels through ensemble learning, while leveraging domain-invariant and domain-specific features for classification.

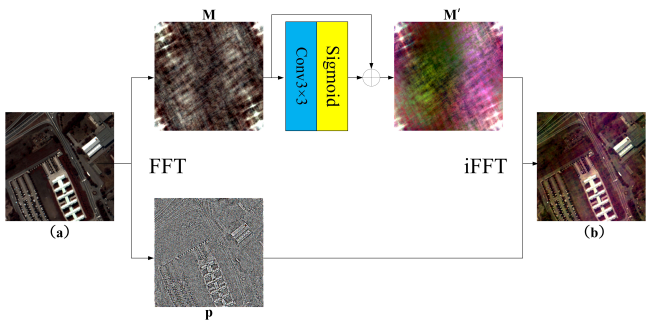

Fig. 3. Schematic of FSAM. Perform a spatial Fourier transform on the HSI and then separate the magnitude and phase. (a) Represents the original HSI, *M* denotes the image reconstructed solely from magnitude, *M'* represents the image reconstructed from the perturbed magnitude, *P* denotes the image reconstructed solely from phase and (b) represents the HSI after Fourier enhancement.

Feature Extraction module, a domain-invariant representation learning module, a domain-specific representation learning module, and an ensemble learning module.

A. Generator

1) Spatial Feature Enhancement Module: Existing DG methods often apply random transformations to spatial features to enhance diversity when generating EDs. However, this approach inevitably results in the loss of domain-invariant information. As indicated in [25] for natural images, the phase component encodes semantic information (e.g., structural textures), while the magnitude component captures style-related statistical characteristics (e.g., lighting and noise). Inspired by this theory, we propose that in the HSI domain, a similar principle can be leveraged to balance invariance and diversity in spatial feature generation. As shown in Fig. 3, reconstructing an image using only the phase component preserves cross-domain invariant structural textures, while reconstruction using only the magnitude component retains domain-specific style attributes. By preserving the phase component and perturbing the magnitude component, the reconstructed image maintains the same structural texture while introducing controllable style variations. The FSAM ensures semantic consistency by retaining the phase component while enhancing feature diversity through controlled magnitude perturbations. This approach effectively balances invariance and diversity in the ED, thereby improving cross-DG.

The Spatial Feature Enhancement module operates as follows: the original features $\mathbf{X}$ first pass through an encoder consisting of Conv2D-BN-LeakyReLU-Conv2D layers to extract shallow spatial features $\mathbf{X}'$. FSAM then processes the shallow spatial features $\mathbf{X}'$. After applying a Fast Fourier Transform (FFT), the features are decomposed into magnitude and phase components. Since phase components retain high-level semantic information, while magnitude components capture low-level statistical attributes, we apply a transformation to the magnitude components to better preserve cross-domain invariant semantic information. Specifically, we pass the magnitude components through a Conv2D-Sigmoid layer before adding them back to the original magnitude components, yielding the transformed magnitude components. Finally, these transformed magnitude components are recombined with the original phase components and processed through an inverse Fast Fourier Transform (iFFT) to generate the enhanced spatial features $\mathbf{X}''$.

To prevent the transformed spatial features from diverging excessively from the original features due to magnitude transformation, we introduce a learnable weight $\mathbf{b}$ to perform

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a weighted sum of the transformed and original features, thereby obtaining the final spatially enhanced feature $\mathbf{X_{spa}}$. This process can be formulated as follows:

$$\mathbf{X_{spa}} = \boldsymbol{b}\mathbf{X}'' + (1-\boldsymbol{b})\mathbf{X}'. \tag{5}$$

2) Spectral Feature Enhancement Module: The spectral features of HSIs are high-dimensional. Current DG methods that directly apply random style transformations to the extracted spectral features can introduce noise due to spectral redundancy, resulting in the ED that differs significantly from the SD. To address this issue, we treat the spectral bands as channels and introduce RSCAM, as shown in Fig. 1, to better focus on valuable spectral bands while suppressing less important ones. In the Spectral Feature Enhancement module, the original features $\mathbf{X}$ are first processed by the RSCAM, which incorporates the CAM to suppress less informative channels, thereby obtaining a refined feature $\mathbf{X}^1$. Next, the RSCAM adaptively fuses the CAM-processed features $\mathbf{X}^1$ with the original features $\mathbf{X}$ using a learnable weight $\boldsymbol{\alpha}$, a process formulated as

$$\hat{\mathbf{X}} = \boldsymbol{\alpha}\mathbf{X}^1 + (1-\boldsymbol{\alpha})\mathbf{X}. \tag{6}$$

Finally, we apply AdaIN [36] to the fused feature $\hat{\mathbf{X}}$ to further enhance spectral adaptability. This can be represented as

$$\text{AdaIN}\left(\hat{\mathbf{X}}, Z\right) = FC_1(Z)\frac{\hat{\mathbf{X}} - \mu\left(\hat{\mathbf{X}}\right)}{\sigma\left(\hat{\mathbf{X}}\right)} + FC_2(Z) \tag{7}$$

where $Z \in \mathbb{R}^{H\times W\times B} \sim$ Normal(0,1) is random noise sampled from a normal distribution, FC_1 and FC_2 denote fully connected layers, and $\mu(\hat{\mathbf{X}})$ and $\sigma(\hat{\mathbf{X}})$ are the mean and standard deviation of the fusion features $\hat{\mathbf{X}}$, respectively.

The features output by AdaIN pass through a batch normalization (BN) layer and a nonlinear activation function (LeakyReLU), and then through a decoder to obtain the spectrally enhanced features $\mathbf{X_{spe}}$.

3) Feature Fusion Module: The spatially enhanced features $\mathbf{X_{spa}}$, spectrally enhanced features $\mathbf{X_{spe}}$, and original spectral patches $\mathbf{X}$ are adaptively fused through learnable weights $\boldsymbol{\beta}_1, \boldsymbol{\beta}_2$, and $\boldsymbol{\beta}_3$ to obtain the generated ED, which can be formalized as

$$\boldsymbol{X_e} = \boldsymbol{\beta}_1\boldsymbol{X_{spa}} + \boldsymbol{\beta}_2\boldsymbol{X_{spe}} + \boldsymbol{\beta}_3\boldsymbol{X} \tag{8}$$

where $\boldsymbol{\beta}_1 + \boldsymbol{\beta}_2 + \boldsymbol{\beta}_3 = 1$.

By simultaneously enhancing spatial and spectral features and incorporating adversarial training, the ED achieves a balance between feature diversity and invariance. Consequently, the discriminator can more effectively extract both domain-invariant and domain-specific features from the ED and SD.

B. Discriminator

The discriminator consists of a feature extraction module, a domain-invariant representation learning module, a domain-specific representation learning module, and an ensemble learning module. The domain-invariant representation learning module aims to extract generalizable and transferable features across training domains, while the domain-specific representation learning module captures distinctive characteristics of each SD to enhance generalization. The overall workflow of the discriminator is illustrated in Fig. 2, with detailed descriptions of each module provided below.

1) Feature Extraction Module: The feature extraction module comprises two Conv2D-ReLU layers, a CBAM attention module [31], and a max pooling layer. The CBAM module integrates both the CAM and the Spatial Attention Module (SAM), sequentially applying them to enhance feature representation. This mechanism adaptively emphasizes informative regions while suppressing less relevant ones, thereby improving the discriminative power of the extracted features.

SAM produces two 2-D maps, $\mathbf{F^s_{avg}} \in \mathbb{R}^{1\times H\times W}$ and $\mathbf{F^s_{max}} \in \mathbb{R}^{1\times H\times W}$, by aggregating channel-wise information from the feature maps using both average pooling and max pooling. They are then concatenated and fed into a convolutional layer to generate a 2-D spatial attention map. The computation of spatial attention can be performed as follows:

$$\begin{aligned}\mathbf{M_s}(\mathbf{F}) &= \sigma\left(f^{7\times7}\left(\left[\text{AvgPool}(\mathbf{F}); \text{MaxPool}(\mathbf{F})\right]\right)\right)\\ &= \sigma\left(f^{7\times7}\left(\left[\mathbf{F^s_{avg}}; \mathbf{F^s_{max}}\right]\right)\right)\end{aligned} \tag{9}$$

where σ represents the sigmoid function, and $f^{7\times7}$ denotes a convolutional layer with a kernel size of 7×7.

2) Domain-Invariant Representation Learning Module: In order to address the variations in feature distributions across different SDs, we designed a domain-invariant representation learning module to boost generalization by learning transferable knowledge that is common across domains. Specifically, we aim to minimize the distributional differences between each domain pair D^i and D^j. To achieve this, we introduce supervised contrastive learning [20] to constrain the features extracted by the feature extraction module, thereby learning class-wise domain-invariant representations. This can be formulated as

$$\mathcal{L}_{\text{supcon}} = -\sum_{i=0}^{N}\frac{1}{|P(i)|}\sum_{p\in P(i)}\log\frac{\exp\left(S\left(\mathbf{z}_i, \mathbf{z}_p^+\right)/\tau\right)}{\sum_{a\in A(i)}\exp\left(S\left(\mathbf{z}_i, \mathbf{z}_a^-\right)/\tau\right)} \tag{10}$$

where for each embedding feature $\mathbf{z}_i$ in the minibatch, $P(i)$ and $A(i)$ represent the sets of positive and negative samples, respectively, with $|P(i)|$ denoting the number of positive samples. $\mathbf{z}_p^+$ and $\mathbf{z}_a^-$ refer to one of the positive and negative samples, respectively. During the optimization of the discriminator, features from the same class in SD, ED, and MD are placed in $P(i)$, while those from different classes are placed in $A(i)$. By optimizing the discriminator with $\mathcal{L}_{\text{supcon}}$, we ensure that samples of the same class have closer distances, while samples of different classes have farther distances. This approach facilitates the discriminator's learning of class-specific shared representations from samples within the same class.

3) Domain-Specific Representation Learning Module: While domain-invariant representation learning enables the model to capture cross-domain invariant information, thereby enhancing generalization, certain domain-specific features remain essential for accurate classification. Relying exclusively on domain-invariant features may lead to performance degradation due to the loss of critical information. To address

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this, the domain-specific representation learning module is designed to complement domain-invariant learning by preserving domain-specific features. Specifically, we introduce domain labels to facilitate domain classification. The model's ability to correctly classify domains relies on domain-specific features, thereby guiding the feature extraction module to capture both domain-invariant and domain-specific information. The domain classification process involves feeding the extracted features into a domain classifier C_d, which consists of three linear layers and two ReLU activations.

In the domain classification task, the loss is computed using cross-entropy loss. The calculation of the cross-entropy loss is as follows:

$$\mathcal{L}_{ce}\left(p_i, y_i\right) = -\sum_{c} y_i^c \log p_i^c \tag{11}$$

where y_i is the one-hot encoded label of x_i, c is the index of the categories, and p_i is the predicted probability output obtained from the classifier C_d. Thus, the classification loss for different SDs is specified as

$$\mathcal{L}_{\text{domain}}\left(X, D\right) = \frac{1}{N}\sum_{i} \mathcal{L}_{ce}\left(C_d\left(x_i\right), d_i\right) \tag{12}$$

where D represents the SD labels, and N denotes the number of SDs.

4) Ensemble Learning Module: This work introduces ensemble learning into DG for HSI classification for the first time, leveraging multiple SDs to infer the TD. Our key assumption is that although test data remain inaccessible during training, they can be approximated as a combination of different training data. This assumption is reasonable since HSIs from various sources often share intrinsic similarities. For example, the reflectance of the same object category across certain spectral bands tends to be consistent, and the spatial relationships among different object categories may exhibit structural similarities. Therefore, it is feasible to represent any TD as a weighted combination of the SDs used during training.

In the ensemble learning module, the goal is to integrate features from multiple SDs into a unified representation. Specifically, we build a base learner for each SD, which is collectively trained to form a stable model that performs consistently well. Each base learner consists of a shared feature extractor and a domain-specific feature extraction layer f_k for each domain D_k. This division is motivated by studies on deep network transfer learning [37], suggesting that lower layers generally learn basic and universal features, whereas higher layers learn domain-specific features. Formally, given new test data $\mathbf{x}$, its feature $\mathbf{z}$ is formulated as

$$\mathbf{z} = \sum_{k=1}^{K} w_k f_k\left(f_e\left(\mathbf{x}\right)\right), \text{where } w_k > 0 \text{ and } \sum_{k=1}^{K} w_k = 1 \tag{13}$$

where f_k denotes the domain-specific feature extraction function for domain D_k, while f_e represents the shared feature extraction function, which corresponds to the feature extraction module. The weights w_k represent the weight on domain D_k, indicating the similarity between the test data and the data in domain D_k. The w_k are computed using (12), where the domain classifier C_d takes the features extracted by the f_e as input and uses SoftMax to satisfy (12), resulting in the probabilities that the test data belongs to each D_k. These probabilities are then taken as the weights w_k.

During testing of the target data, we do not have access to the domain labels of unseen TDs. The shallow features extracted by the feature extractor are passed through the domain classifier, and its output is used as the weight to fuse the deep features extracted after passing through each domain-specific feature extractor. Essentially, the domain-specific branches, which have been learned, adaptively combine the domain-specific features extracted from every SD in order to form a feature representation for TD. These features are then fed into the classifier C_L to obtain the final classification results.

C. Model Optimization

The optimization process of FSENet consists of two parts: generator optimization and discriminator optimization. During the training phase, the model initially optimizes the discriminator using SD, ED, and MD data as inputs. The final classification loss still employs cross-entropy loss. The losses for SD, ED, and MD can be represented by the following equations:

$$\mathcal{L}_{\text{SD}}\left(X, Y\right) = \frac{1}{N}\sum_{i} \mathcal{L}_{ce}\left(C_L\left(x_i\right), y_i\right) \tag{14}$$

$$\mathcal{L}_{\text{ED}}\left(X_e, Y\right) = \frac{1}{N}\sum_{i} \mathcal{L}_{ce}\left(C_L\left(x_i\right), y_i\right) \tag{15}$$

$$\mathcal{L}_{\text{MD}}\left(X_m, Y\right) = \frac{1}{N}\sum_{i} \mathcal{L}_{ce}\left(C_L\left(x_i\right), y_i\right) \tag{16}$$

where $\mathcal{L}_{ce}$ denotes the cross-entropy loss function, C_L represents the fully connected layer used for classification, x_i is the input data sample, y_i is the corresponding ground truth label, and N is the total number of samples.

The objective of the discriminator's optimization is a weighted combination of (10), (12), and (14)–(16), as follows:

$$\min_{\theta_D} \mathcal{L} = \mathcal{L}_{\text{SD}} + \mathcal{L}_{\text{ED}} + \mathcal{L}_{\text{MD}} + \lambda_1 \mathcal{L}_{\text{supcon}} + \lambda_2 \mathcal{L}_{\text{domain}} \tag{17}$$

where λ_1 and λ_2 are hyperparameters that balance the domain classification loss and the supervised contrastive learning loss.

Through adversarial training, the generator is capable of producing diverse ED. However, without imposing constraints on the generator, the class semantic information of the ED may deviate from that of the SD, hindering the discriminator from capturing class-invariant features across domains. To address this issue, the class semantic information between ED and SD is constrained using an MMIRN [22], which maximizes the MI between ED and the SD. Inspired by contrastive learning [38], the lower bound of the MI between ED and SD [39] can be estimated using the InfoNCE loss [40], which can be formalized as follows:

$$\mathcal{L}_{\text{MI}}\left(X, X_e\right) = \log \frac{\exp\left(X_e \cdot X/\tau\right)}{\sum_{X^- \in N(i)} \exp\left(X \cdot X^-/\tau\right)} \tag{18}$$

where τ denotes the temperature parameter, and $N(i)$ represents the set of negative samples. The $\mathcal{L}_{\text{MI}}(X, X_e)$ loss is

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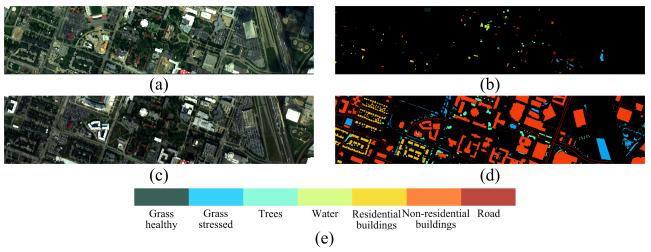

Fig. 4. Pseudo-color images and ground truth for the Houston 13 and Houston 18 datasets. (a) Pseudo-color image of Houston 13. (b) Ground truth of Houston 13. (c) Pseudo-color image of Houston 18. (d) Ground truth of Houston 18. (e) Colormap.

designed to maximize the MI between the ED and SD, thereby boosting the semantic consistency between the ED and the SD.

To optimize the generator, (18) is solved, and (12) is employed as an auxiliary loss to ensure that the ED maintains a certain level of discrepancy with the SD. The generator's optimization goal is a weighted combination of (12) and (18) as follows:

$$\min_{\theta_G} \mathcal{L} = \mathcal{L}_{\text{MI}} + \lambda_2 \mathcal{L}_{\text{domain}} \tag{19}$$

where λ_2 is a hyperparameter that balances the domain classification loss.

IV. Experiment Results and Analysis

A. Dataset Description

To evaluate the effectiveness of FSENet in cross-scenario HSI classification under the single-source DG setting, we conducted experiments on four cross-scenario HSI classification datasets: 1) Houston 13 and Houston 18; 2) Pavia University and Pavia Center; 3) Dioni and Loukia; and 4) HongHu and HanChuan. Detailed descriptions of these datasets are provided below.

1) Houston 13 and Houston 18 Datasets: The Houston 13 dataset [41] and the Houston 18 dataset [42] were collected by different hyperspectral sensors over the campus and surrounding areas of the University of Houston in 2012 and 2017, respectively. The Houston 13 dataset comprises 144 spectral bands ranging from 380 to 1050 nm, with a spatial size of 349 × 1905 pixels and a spatial resolution of 2.5 m. The Houston 18 dataset includes 48 spectral bands within the same range of 380–1050 nm, captured by the ITRES CASI 1500 sensor, with a spatial size of 1202 × 4172 pixels and a spatial resolution of 1 m. For the experiments, overlapping regions containing similar land cover types were retained from both the Houston 13 and Houston 18 datasets. The spatial resolution of the Houston 18 dataset was downsampled [43] to match that of the Houston 13 dataset, resulting in an overlapping region of 209 × 955 pixels. Experiments were conducted using the 48 spectral bands corresponding to the Houston 18 dataset on the Houston 13 dataset. The Houston 13 dataset served as the SD, while the Houston 18 dataset served as the TD. Fig. 4 shows the pseudo-color images of the Houston 13 and Houston 18 datasets, which contain seven consistent classes as listed in Table I.

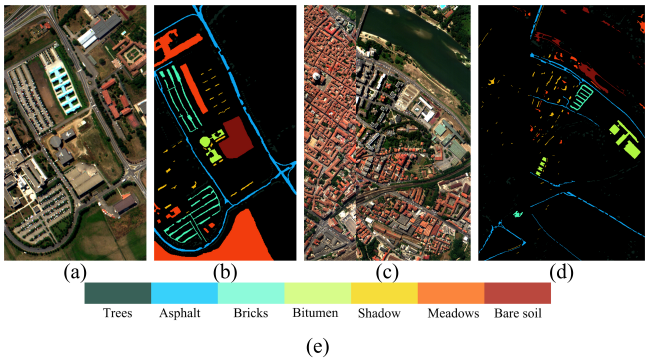

Fig. 5. Pseudo-color images and ground truth for the Pavia University and Pavia Center datasets. (a) Pseudo-color image of Pavia University. (b) Ground truth of Pavia University. (c) Pseudo-color image of Pavia Center. (d) Ground truth of Pavia Center. (e) Colormap.

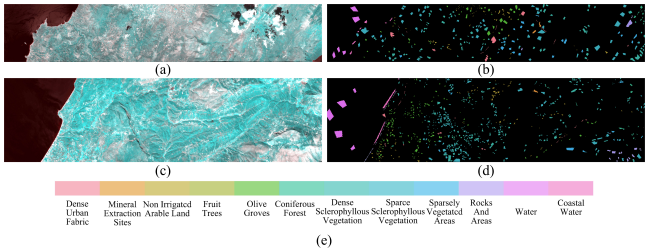

Fig. 6. Pseudo-color images and ground truth for the Dioni and Loukia datasets. (a) Pseudo-color image of Dioni. (b) Ground truth of Dioni. (c) Pseudo-color image of Loukia. (d) Ground truth of Loukia. (e) Colormap.

2) Pavia University and Pavia Center Datasets: The Pavia University and Pavia Center datasets were acquired by the Reflectance Optical Spectral Imaging System (ROSIS) Sensor in northern Italy. The Pavia University dataset has a spatial size of 610 × 340 with 103 bands. The Pavia Center dataset has a spatial size of 1096 × 715 with 102 bands. They both have a spatial resolution of 1.3 m, a spectral range of 430–860 nm, and cover scenes in urban areas. In the experiment, the Pavia University dataset is used as the SD, the Pavia Center dataset is used as the TD, and the last spectral band of the Pavia University dataset is removed so that Pavia University and Pavia Center have the same number of spectral bands. Fig. 5 displays the pseudo-color images of the Pavia University and Pavia Center datasets, along with their ground truth. These datasets include seven consistent classes, as outlined in Table II.

3) Dioni and Loukia Datasets: The dataset covers two regions, Dioni and Loukia [44]. The Dioni dataset has dimensions of 250 × 1376, and the Loukia dataset has dimensions of 249 × 945. Both regions have a spatial resolution of 30 m. The HSI, captured by the Hyperion sensor, includes 176 spectral bands and 14 identical categories. In this article, we use 176 spectral bands and 12 object categories. Fig. 6 shows the pseudo-color images of the Dioni and Loukia datasets with their ground truth, and Table III details the number of samples.

4) HongHu and HanChuan Datasets: Both HongHu and HanChuan datasets were collected using the Headwall Nano-Hyperspec imaging sensor covers a spectral range from

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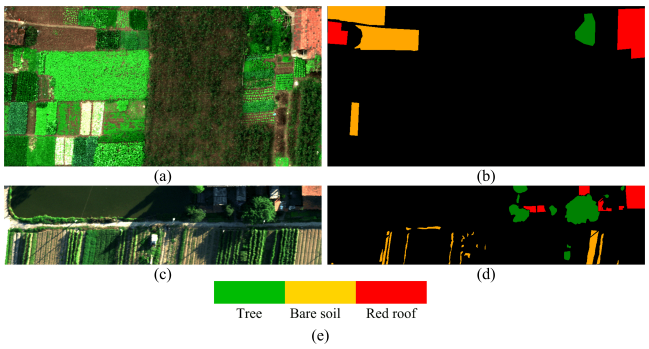

Fig. 7. Pseudo-color images and ground truth for the HongHu and HanChuan datasets. (a) Pseudo-color image of HongHu. (b) Ground truth of HongHu. (c) Pseudo-color image of HanChuan. (d) Ground truth of HanChuan. (e) Colormap.

400 to 1000 nm. HongHu dataset has dimensions of 940 × 475 pixels, contains 270 bands, and has a spatial resolution of 0.043 m. HanChuan dataset has dimensions of 1217 × 303 pixels, contains 274 bands, and has a spatial resolution of 0.109 m. For this experiment, 270 spectral bands of HanChuan dataset were selected as input, focusing on three experimental classes: trees, red rooftops, and bare soil. The number of samples for each class is shown in Table IV, and Fig. 7 displays a false-color image with a real background.

B. Experimental Setup

1) Comparison Methods: To evaluate the effectiveness of FSENet in cross-scene classification, we conducted experiments on the Houston 18, Pavia Center, Loukia, and HanChuan datasets. Eight cross-scene classification methods were selected for comparison, including three DA-based methods and five single-source DG-based methods. The DA-based methods included Domain-Sample Alignment Network (DSAN) [45], Multirepresentation Adaptation Network (MRAN) [46], and Two-Stage Transfer Network (TSTnet) [47]. The DG-based methods included Single-Domain Extension Network (SDEnet) [20], LLURnet [21], Spectral–Spatial Adversarial Multidomain Synthesis Network (S2AMSnet) [22], Spectral–Spatial Enhancement and Causal Constraint Network (S2ECnet) [23], and Frequency Disentanglement-based DG Network (FDGnet) [24].

For the DA-based methods, 80% of the labeled data from the SD, along with their corresponding labels, were used for training. Additionally, all data from the TD were included in the training process. The remaining 20% of the labeled SD data were set aside as a validation set to assess model performance. For the single-source DG-based methods, only 80% of the labeled SD data and their corresponding labels were used for model training, while the remaining 20% of the labeled SD data were reserved as the validation set.

For the DSAN method, ResNet-18 was employed as the feature extractor. To meet the input size requirements of ResNet-18, spectral patches of size 32 × 32 were used as inputs. The parameter configurations for MRAN, TSTnet, SDEnet, LLURnet, S2AMSnet, S2ECnet, and FDGnet followed the optimal settings reported in their respective original papers.

2) Evaluation Criteria: To evaluate the cross-scenario HSI classification performance of FSENet compared to the eight methods, qualitative analysis based on classification maps was conducted. Additionally, class-wise accuracy (CA), OA, and the KC were utilized as quantitative evaluation metrics to quantitatively assess the performance of the proposed method relative to the comparative approaches.

3) Experimental Settings:

a) Data preprocessing: To address the differences in the number of spectral bands between the SD and TD, we employed various methods to align their band counts. Specifically, for the Houston 13 dataset, which originally has 144 spectral bands, we selected the 48 bands corresponding to those in the Houston 18 dataset for the experiments. For the Pavia University dataset, we removed its last spectral band to match the 102 bands of the Pavia Center dataset. No adjustments were necessary for the Dioni and Loukia datasets, as they already share the same number of spectral bands. For the HanChuan dataset, we removed its last four bands to align with the 270 bands of the HongHu dataset. To mitigate the differences in data distribution caused by the varying scales of the SD and TD, and to enhance model stability, we normalized the data from both domains before training and testing. Additionally, we augmented the training data with techniques including random flipping, radiation noise, and mixture noise. Furthermore, based on the patch_size, we applied edge padding to the training data to ensure consistency.

b) Training process: Similar to the comparative methods, we divide the training samples into an 80% training set and a 20% validation set. During training, we first load the training set and perform data preprocessing. The training set is then fed into the generator to generate an ED and a MD. Subsequently, these three domains are input into the discriminator for training. After calculating the discriminator loss, the discriminator parameters are updated. Next, the generator is trained by calculating the generator loss, and the generator parameters are updated. The two components are trained alternately. Finally, the classification performance of the discriminator is evaluated on the validation set, and the best model and evaluation results are saved.

c) Hyperparameter settings: In the four cross-scene HSI classification tasks, the learning rate for FSENet was set to $1e^{-4}$, the batch size was set to 256, the number of training epochs was set to 400, and the random seed was set to 333. For the spectral patch size (patch_size) and the loss function weights (lambda_1 and lambda_2), different settings were applied to the four datasets. For the Houston dataset, the patch_size was set to 13 × 13, and both lambda_1 and lambda_2 were set to 1.0. For the Pavia dataset, the patch_size was set to 9 × 9, and both lambda_1 and lambda_2 were set to 1.0. For the HyRANK dataset, the patch_size was set to 7 × 7, with lambda_1 and lambda_2 set to 0.01 and 10, respectively. For the WHU dataset, the patch_size was set to 11 × 11, and both lambda_1 and lambda_2 were set to 1.0. (The values of

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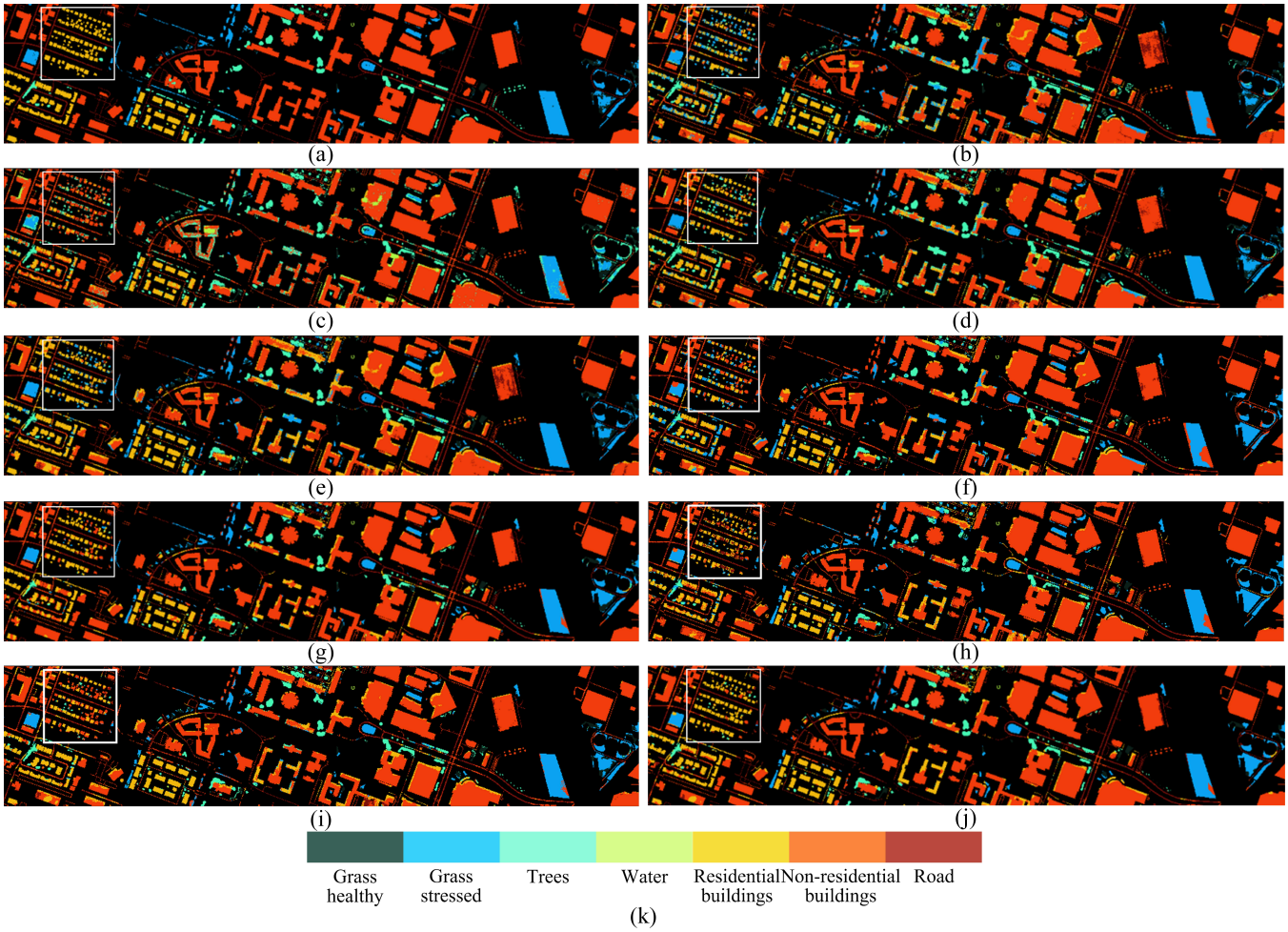

Fig. 8. Classification maps of FSENet and comparative methods on the Houston 18 target dataset. (a) Ground truth. (b) DSAN. (c) MRAN. (d) TSTnet. (e) SDEnet. (f) LLURnet. (g) S2AMSnet. (h) S2ECnet. (i) FDGnet. (j) FSENet. (k) Colormap.

patch_size, lambda_1, and lambda_2 were determined through parameter analysis.)

Other fusion weights in the model were defined as self-learning parameters and were automatically learned by the model. Additionally, to evaluate the reliability of the algorithms, each cross-scenario HSI classification task involved conducting ten repetitions of the experiments for both the proposed method and the comparative methods. The results were then averaged to assess the performance of the methods.

d) Experimental environment: The experimental hardware setup consisted of an Intel Core i7-14900K 24-core CPU processor and an NVIDIA GeForce RTX 1080 Ti GPU with 10 GB of RAM. The proposed method and the comparative methods were implemented using Python 3.10.14 and PyTorch 1.11.

C. Results and Discussion

First, a qualitative visual evaluation of the target scene classification maps obtained by FSENet and the eight comparative methods was performed. Figs. 8–11 present the classification maps of FSENet and the comparative methods on Houston 18, Pavia Center, Loukia, and HanChuan, respectively. Unlabeled pixels are represented as background, while labeled pixels indicate predicted classes. The classification maps produced by FSENet more closely resemble the ground truth of all target scenes.

To further quantitatively evaluate the performance of FSENet against competing methods, Tables V–VIII report the CA, OA, and KC achieved by FSENet and the eight comparative methods on Houston 18, Pavia Center, Loukia, and HanChuan, respectively. Based on these quantitative evaluation metrics, it can be concluded that among the DA-based methods, TSTnet achieved the highest OA on the Houston 18 target scene, while among the five comparative single-source DG methods, S2AMSNet attained the highest OA. On the Pavia Center target scene, DSAN outperformed MRAN and TSTnet among the DA-based methods, whereas S2AMSNet demonstrated superior performance among the single-source DG methods. On the Loukia target scene, TSTnet performed the best among the DA-based methods, while FDGnet exhibited the optimal performance among the single-source DG methods. On the HanChuan target scene, TSTnet showed the best performance among the DA-based methods, and S2AMSnet achieved the best results among the single-source DG methods. Notably, in all four target scenes, the optimal single-source DG-based methods outperformed the best

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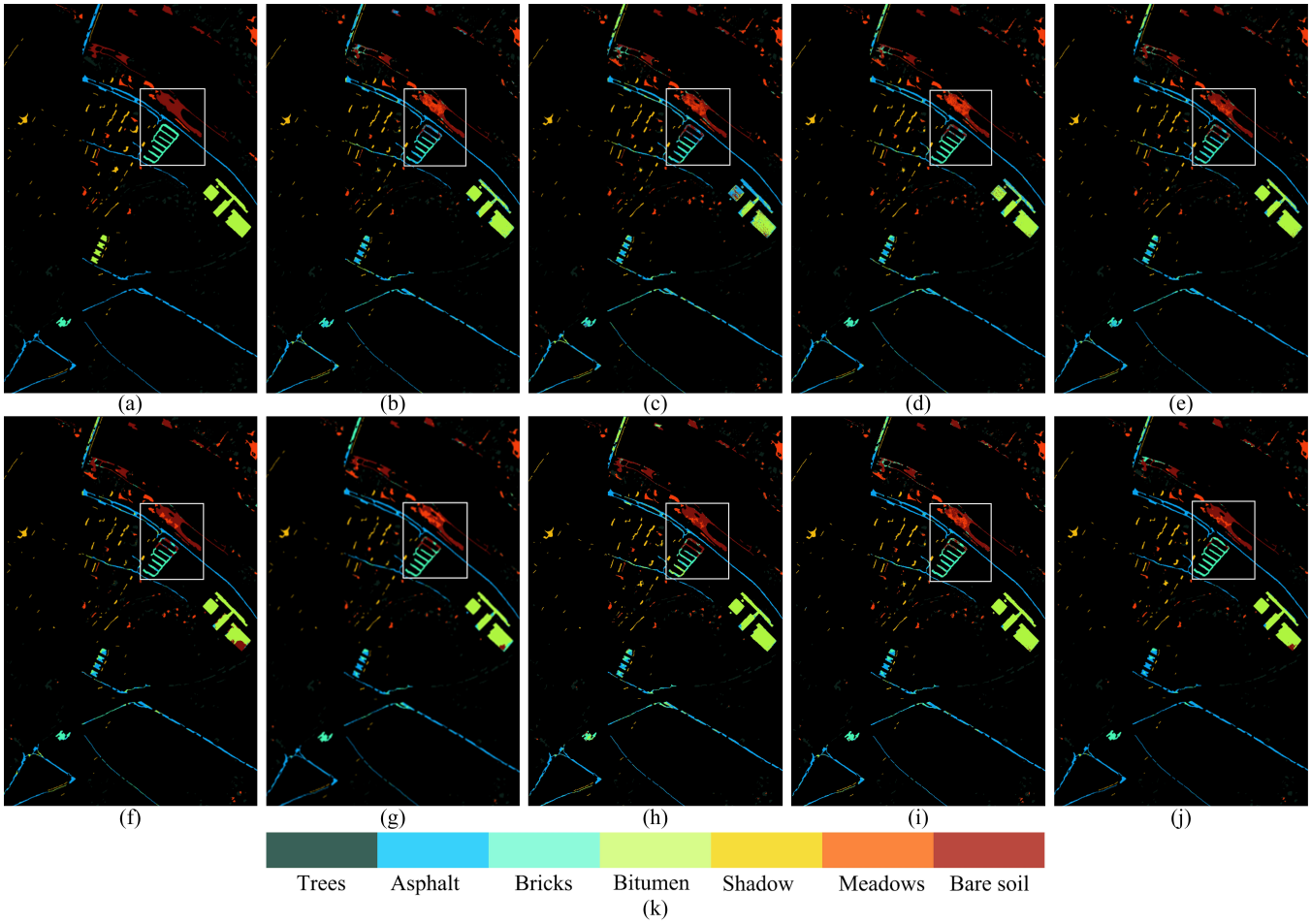

Fig. 9. Classification maps of FSENet and comparative methods on the Pavia Center target dataset. (a) Ground truth. (b) DSAN. (c) MRAN. (d) TSTnet. (e) SDEnet. (f) LLURnet. (g) S2AMSnet. (h) S2ECnet. (i) FDGnet. (j) FSENet. (k) Colormap.

DA-based methods. This indicates that DA methods, which rely on domain alignment strategies to reduce inter-domain shifts, have limitations in effectively generalizing to scenarios with complex domain variations. In contrast, single-source DG methods enhance the model's generalization capability by expanding domain diversity according to the characteristics of HSIs, demonstrating superior performance in handling complex domain variations without leveraging TD data.

The proposed FSENet, designed for single-source DG in cross-scenario HSI classification, exhibited competitive performance across all target scenes in our experiments. As shown in Table V, compared to the best DA method and the best single-source DG method, FSENet improved the OA on Houston 18 by 7.89% and 2.41%, respectively. The white boxes in Fig. 8 highlight that FSENet achieved significant improvements in class five (residential buildings). Similarly, as presented in Table VI, FSENet increased the OA on Pavia Center by 8.84% and 4.74%, respectively. The white boxes in Fig. 9 indicate substantial improvements in class three (bricks) and class seven (bare soil). According to Table VII, FSENet enhanced the OA on Loukia by 4.59% and 4.41%, respectively, with the white boxes in Fig. 10 demonstrating notable improvements in class nine (sparse sclerophyllous vegetation) and class ten (rocks and sand). Finally, as shown in Table VIII, FSENet improved the OA on HanChuan by 12.91% and 2.27%, respectively, with the white boxes in Fig. 11 highlighting significant improvements in class three (red roof).

Compared to the best single-source DG methods, such as S2AMSnet and FDGnet, FSENet adopts a similar generator-discriminator architecture and utilizes the MMIRN from S2AMSnet to constrain the similarity between the ED and the SD. However, FSENet introduces significant improvements in the design of both the generator and the discriminator. While FSENet, S2AMSnet, and FDGnet all involve spatial and spectral feature enhancement, S2AMSnet and FDGnet only perform simple randomization on spatial and spectral features. This approach, also used by other DG methods, may lead to the loss of potential domain-invariant features and neglect the redundancy of spectral bands. In FSENet, we propose the FSAM, which uses amplitude enhancement via Fourier transforms to maximally preserve domain-invariant information. We also design the RSCAM to randomly enhance important spectral bands, thereby avoiding the issue of band redundancy. For the discriminator, unlike S2AMSnet and FDGnet, which rely solely on contrastive learning to enforce domain-invariant

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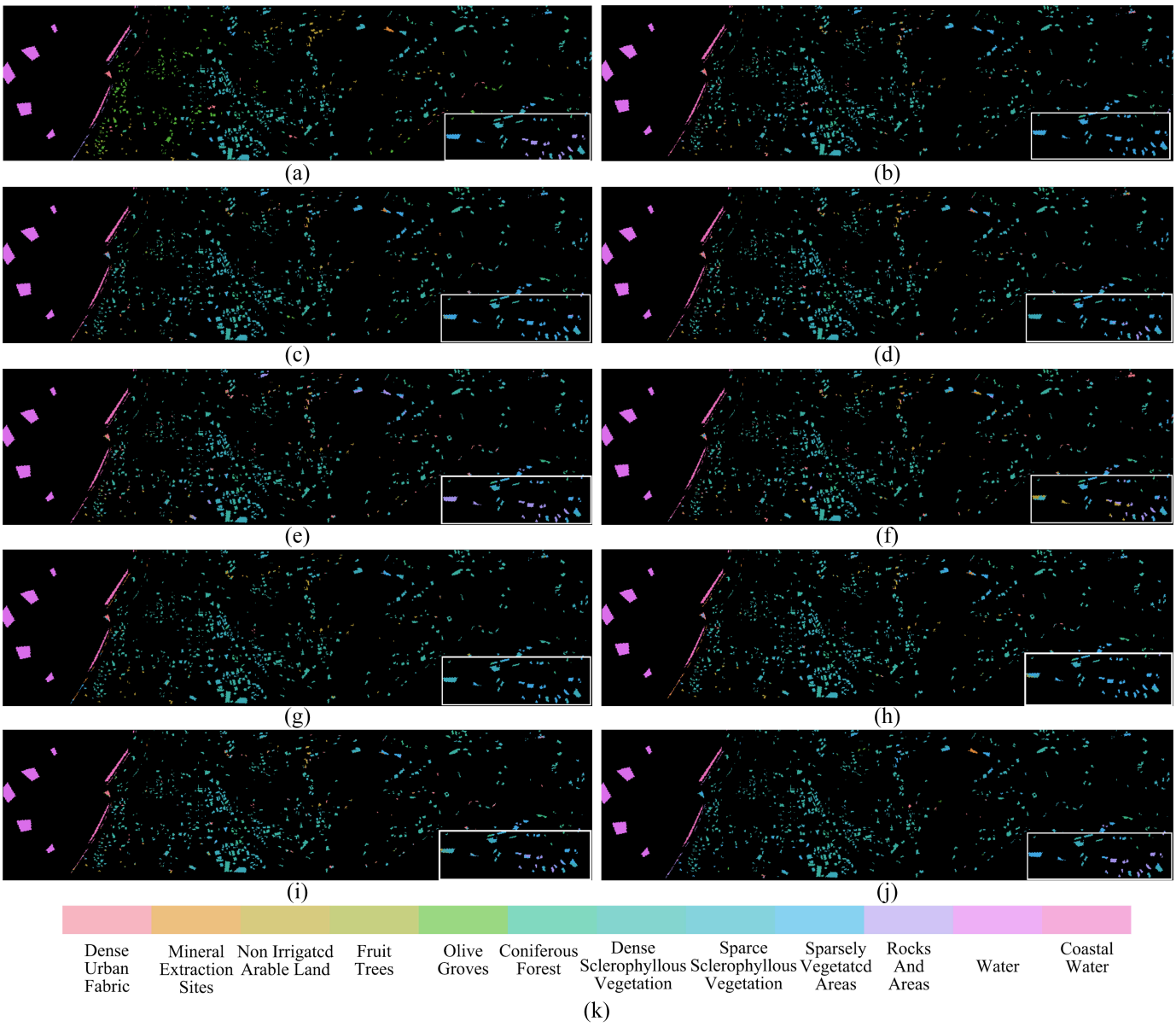

Fig. 10. Classification maps of FSENet and comparative methods on the Loukia target dataset. (a) Ground truth. (b) DSAN. (c) MRAN. (d) TSTnet. (e) SDEnet. (f) LLURnet. (g) S2AMSnet. (h) S2ECnet. (i) FDGnet. (j) FSENet. (k) Colormap.

feature extraction, FSENet not only employs contrastive learning but also introduces the DLEF. This framework incorporates a domain classification loss, ensuring that the model extracts both domain-invariant features beneficial for classification and domain-specific features that capture distinctive domain characteristics. Furthermore, DLEF integrates ensemble learning, enabling the classifier to aggregate features from multiple SDs, thereby generating more discriminative representations for the final classification. The improvements of 2.41%, 4.74%, 4.41%, and 2.27% in OA on the four target scenes of Houston 18, Pavia Center, Loukia, and HanChuan demonstrate that the FSAM and RSCAM proposed in FSENet can effectively generate different EDs from spatial and spectral dimensions. Furthermore, the discriminator employing DLEF can better extract domain-invariant and domain-specific features for final classification, successfully fusing features from multiple SDs to achieve robust generalization performance in the TD.

D. Parameter Analysis

A parameter sensitivity analysis was performed to assess FSENet's sensitivity to four TDs. Similar to all single-source DG methods, we investigated the impact of the input spectral patch size (patch_size) on classification accuracy. The results are shown in Fig. 12. The optimal patch_size values for the four target scenes—Houston 18, Pavia Center, Loukia, and HanChuan—were 13, 9, 7, and 11, respectively. The regularization parameters λ_1 and λ_2, considered as tunable hyperparameters, were selected from the set {1e+2, 1e+1, 1e+0, 1e-1, 1e-2}. λ_1 and λ_2 are critical hyperparameters in FSENet, as they not only control the contribution of supervised

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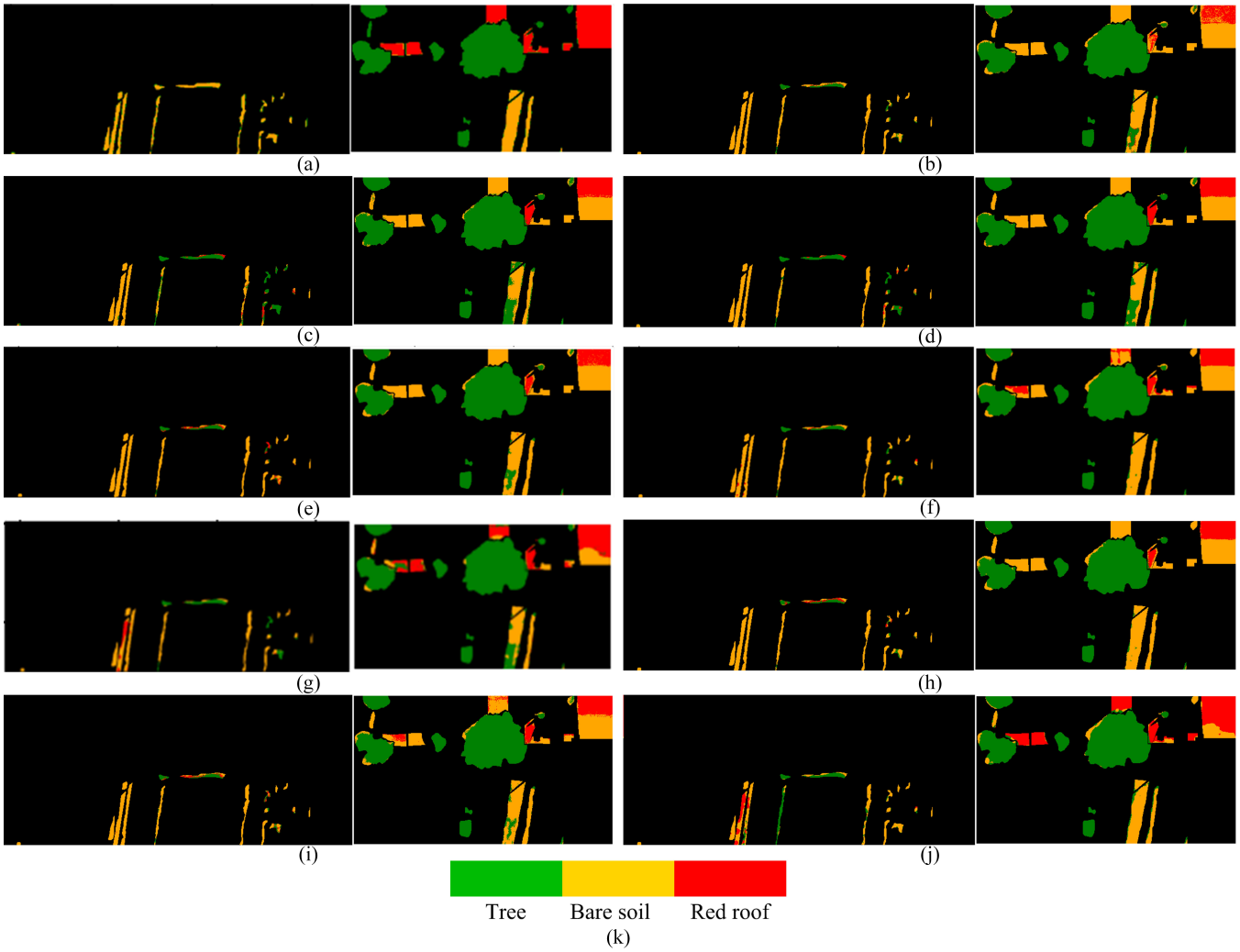

Fig. 11. Classification maps of FSENet and comparative methods on the HanChuan target dataset. (a) Ground truth. (b) DSAN. (c) MRAN. (d) TSTnet. (e) SDEnet. (f) LLURnet. (g) S2AMSnet. (h) S2ECnet. (i) FDGnet. (j) FSENet. (k) Colormap.

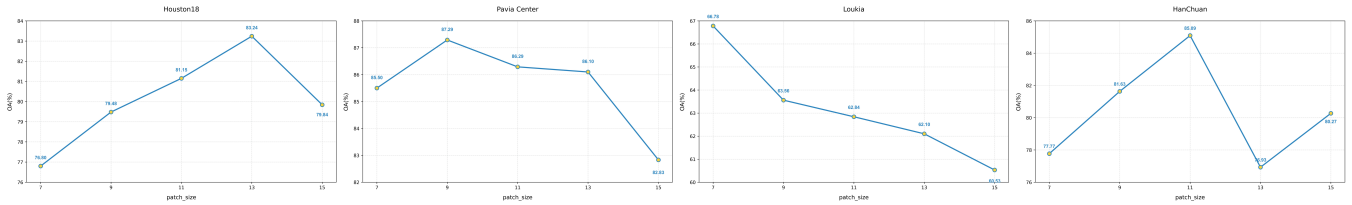

Fig. 12. Parameter tuning of patch_size for the proposed FSENet using all the four experimental data.

contrastive learning and adversarial learning to DG but also regulate the impact of domain-invariant and domain-specific features on DG. As shown in Fig. 13, for the Houston 18, Pavia Center, and HanChuan target scenes, the optimal values for λ_1 and λ_2 are both 1e+0. For the Loukia target scene, the optimal values for λ_1 and λ_2 are 1e-2 and 1e+1, respectively.

E. Computational Complexity

We also evaluated the running time of all participating experimental methods, as shown in Table IX. The reported results represent the time required for a single epoch. Across all four experimental settings, the proposed method demonstrated the shortest running time. This is primarily due to the fact that many comparative methods utilize a greater number of network layers and adopt deeper architectures. However, experimental results indicate that reducing the number of network layers does not significantly impact classification accuracy.

F. Ablation Studies

To generate spatially enhanced features that retain as much domain-invariant information as possible, we proposed the FSAM, which employs amplitude transformation augmentation techniques. To mitigate the noise effect caused

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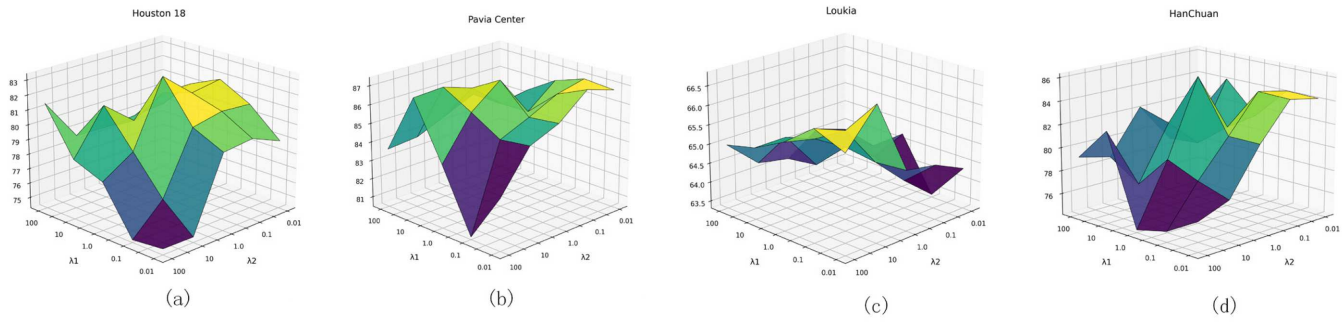

Fig. 13. Parameter tuning of λ_1 and λ_2 for the proposed FSENet using all the three experimental data. (a) Houston 18. (b) Pavia center. (c) Loukia. (d) HanChuan.

TABLE IX
TRAINING TIME (IN SECONDS) OF DIFFERENT METHODS ON FOUR DATASETS. THE SHORTEST ONE IS SHOWN IN BOLD

Dataset	DSAN[44]	MRAN[45]	TSTnet[46]	SDEnet[20]	LLURnet[21]	S2AMSnet[22]	S2ECnet[23]	FDGnet[24]	FSENet
Houston	16.64	13.52	6.5	6.35	15.03	14.87	162.87	7.23	**3.33**
Pavia	31.76	29.01	34.88	27.07	41.07	57.62	283.01	35.32	**24.04**
HyRANK	23.58	20.06	22.35	20.63	29.52	35.87	156.04	28.29	**11.43**
WHU	71.64	62.48	59.28	56.83	61.49	94.74	309.81	85.36	**50.92**

TABLE X
ABLATION STUDY OF THE FSAM, RSCAM, AND DLEF MODULES OF FSENET ON THREE CROSS-SCENE HSI CLASSIFICATION TASKS (MEAN ± STANDARD DEVIATION)

FSENet			Target Scene OA(%)			
FSAM	RSCAM	DLEF	Houston18	PaviaCenter	Loukia	HanChuan
×	√	√	82.48±0.47	87.05±0.73	65.96±0.63	83.26±3.18
√	×	√	82.84±0.38	86.23±0.83	65.99±0.64	83.95±1.77
√	√	×	82.16±0.64	85.99±0.98	64.98±0.35	84.16±2.49
√	√	√	**83.24**±0.46	**87.29**±0.33	**66.78**±0.50	**85.09**±1.97

by potential spectral band redundancy, we designed the RSCAM to select valuable spectral bands for feature enhancement. Additionally, to enable the discriminator to extract more discriminative features for classification, we introduced the DLEF to aggregate features from multiple SDs.

We conducted ablation studies by individually removing the FSAM, RSCAM, and DLEF to evaluate the effectiveness of each proposed component. The results of these studies, performed on four cross-scenario HSI classification tasks, are reported in Table X. Each ablation experiment was repeated ten times, and the average results were presented to minimize the impact of randomness. The observed decrease in OA when any single module was removed—compared to the network's performance with all modules intact—across the four target scenes further validates the effectiveness of the proposed network architecture.

V. CONCLUSION

This article proposes the FSENet for single-source DG to address domain shifts in cross-scenario HSI classification. By integrating the FSAM and the Random-augmented Spectral Channel Attention Mechanism, FSENet introduces a novel synergistic design that combines Fourier enhancement with dynamic spectral band optimization, effectively mitigating feature distortion and redundant noise—two key bottlenecks in existing methods. The DLEF jointly optimizes domain classification loss and contrastive learning, effectively fusing domain-invariant features (cross-scenario robustness) and domain-specific features (scene discriminability). Experiments on four cross-scenario datasets demonstrate that FSENet achieves an OA improvement of 2.27%–4.74% over state-of-the-art methods, with ablation studies confirming the individual contributions of each module.

However, the proposed framework has the following limitations, and our future experiments will quantify the performance impact of these limitations.

1) *Sensitivity to Integration Configurations:* The current integration architecture relies on weights derived from domain classification for feature fusion. However, research on alternative integration configurations remains limited, including the timing of feature fusion (e.g., early or late fusion) and the use of multiple classifiers for integrating classification results.

2) *Limited Cross-Sensor Generalization:* Current validation is restricted to single-sensor scenarios and does not consider cross-sensor challenges, such as feature mismatches caused by differences in spectral resolution.

Future research will focus on the following.

1) A Reinforcement Learning (e.g., Q-learning) or DARTS-Based Framework to automatically determine optimal feature fusion timing (early/late) and classifier weight associations, including Q-learning optimizations for FSAM's perturbation strategy to enhance feature diversity.

2) Cross-Sensor Unified Representation Learning: This involves combining physical spectral correction models (e.g., SEN2COR) with deep feature disentanglement techniques

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to develop multisensor compatible, domain-invariant feature extraction modules.

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